Venkata Lakshmi Parimala Pasupuleti

Software Engineer · Python · SQL · ETL/ELT Pipelines · AWS S3 · Large-Scale Data Processing · Production Support

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Virginia, USA · Available for DC onsite · Authorized to work in the US (F-1 OPT)

PROFESSIONAL SUMMARY

Software Engineer with **8+ years of experience** building production **data pipelines, ETL/ELT processes**, and large-scale **data extraction, transformation, and loading** workflows in enterprise software environments. At Entrata, owned the annual HUD income-limit **ETL pipeline** end-to-end for nearly 7 years - a few thousand federal source rows transformed into **~180,000 domain records** powering eligibility, rent, and compliance calculations for 100K+ housing units. Deep **SQL development** expertise (200+ queries optimized, 30% performance gain), **Python scripting** for data processing (Pandas, NumPy, PL/SQL), **AWS S3** cloud data integration, and strong **production support** and troubleshooting background. Experienced working directly with stakeholders to analyze **business requirements** and translate them into reliable data management systems. M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026).

CORE COMPETENCIES

Python ETL Development | SQL Development | ETL/ELT Pipeline Design | Data Extraction, Transformation & Loading | Large Datasets & Large-Scale Data Processing | Python Scripting (Pandas, NumPy) | Reference Data Management | Cloud Data Integration (AWS S3) | Data Management Systems | Production Support & Troubleshooting | Business Requirements Analysis | Power BI Reporting | Stored Procedures & PL/SQL | Stakeholder Communication

EXPERIENCE

Entrata (PropTech / Real Estate SaaS)

Sep 2018 – Aug 2024

Software Engineer · SQL · Python · PostgreSQL · AWS S3 · RabbitMQ

- **Owned the annual HUD income-limit ETL pipeline for nearly 7 years** - designed and maintained the full **data extraction, transformation, and loading** workflow that processed a few thousand federal source rows into **~180,000 domain records** powering eligibility, rent, HAP, and tax-credit calculations for 100K+ housing units; every release included source validation, staging, reconciliation, and rollback preparation - zero compliance violations across the ownership period.
- **Built and maintained Python ETL/ELT pipelines** processing **1.6M+ records** across HUD and Tax Credit data sources (5 years of historical data); designed generic **SQL scripts and stored procedures** for **data extraction** from internal and external sources, structured transformation, effective dating, and publication to downstream modules.
- **Optimized 200+ SQL queries** (30% performance gain) through execution plan analysis, indexing, partitioning, and **stored procedure** refactoring across complex multi-tenant **data management systems** - maintained schemas for eligibility, financial calculations, and document workflows; debugged production data issues at the query and data model level.
- **Led cloud data integration to AWS S3** - designed **data loading** workflows sourcing document and resident data from internal systems into cloud storage with Libsodium-encrypted pipelines; achieved 100% PHI compliance (zero security incidents); resolved 90% of legacy storage constraints.
- **Automated Python data processing pipelines** eliminating 80% of manual operational workload; built **reference data management** workflows for income limits, AMI thresholds, and eligibility definitions used across 20+ product modules; collaborated directly with stakeholders and business analysts to **analyze requirements** and translate them into data models and ETL processes.
- Provided **production support and troubleshooting** for data pipeline failures, data integrity issues, and downstream calculation errors; built internal debugging tools reducing debugging time by 40%; maintained bug count below 10 for 6+ consecutive months; delivered **Power BI** and dashboard-based reporting for operational and compliance data.

Entrata India (Db Xento Systems)

Jan 2017 – Sep 2018

Associate Software Engineer · Python · SQL · PostgreSQL · AWS S3

- Built **ETL workflows** for property management and affordable housing data: **income limits import** (1.6M HUD/Tax Credit records across 5 years), document management, and compliance data pipelines using Python, PostgreSQL, and PL/SQL.
- Led **cloud data migration** of thousands of client documents from legacy server storage to **AWS S3**; redesigned document storage database architecture, migrating hundreds of thousands of records to a modern schema with 6 months of post-deployment production support.

George Mason University — Research Associate

Aug 2024 – Aug 2025

Research Associate · Python · R · Statistical Analysis · Power BI

- Built multi-source **Python and R data processing pipelines** on WHO and UK Road Safety datasets (190+ countries) - **data extraction** from external sources, transformation, validation, and statistical modeling achieving $R^2 = 0.977$ regression accuracy.
- Designed automated **data integrity** checks using Isolation Forest to surface anomalies across 3 research pipelines before reporting; built **Power BI** and Tableau dashboards for faculty leadership stakeholders.

PORTFOLIO PROJECTS

CrashClarity — Large-Scale Data Processing Pipeline (104K Records)

Python ETL SQL

2025 · Python, R, scikit-learn, Pandas, NumPy, Tableau · View Project

- End-to-end **data extraction, transformation, and loading** pipeline on 104K+ UK transportation records - multi-source data collection, cleaning, feature engineering across behavioral and environmental variables; built interactive Tableau dashboard for stakeholder reporting; 85%+ model accuracy.

Leasel — AI-Powered Data Investigation Platform

Python FastAPI ETL

2026 · Python, FastAPI, LangGraph, SQLite, TF-IDF · GitHub

- Built **data extraction and transformation** pipelines ingesting regulatory documents, historical incident records, and live system data into a unified **data management** layer; designed structured **data loading** workflows from multiple internal and external sources into SQLite with full lineage and audit logging; 93.6% precision across 1,240 compliance records.

EDUCATION

M.S. Data Analytics Engineering — George Mason University, Virginia — May 2026 · GPA 3.87

Coursework: Data Engineering, ETL Systems, Database Systems, Statistical Analysis, Data Mining, Business Analytics, Cloud Computing

B.Tech Computer Science & Engineering — JNTUA, India — 2013–2017

TECHNICAL SKILLS

ETL & Data Processing: Python ETL Development, ETL/ELT Pipeline Design, Data Extraction Transformation Loading, Python Scripting, Pandas, NumPy, PL/SQL, Stored Procedures, Data Validation, Reference Data Management, Data Management Systems

SQL & Databases: SQL Development, PostgreSQL, MySQL, Oracle, Query Optimization, Execution Plan Analysis, Indexing, Partitioning, Audit Tables, Schema Design, Data Modeling

Cloud & Integration: AWS S3, AWS RDS, Cloud Data Integration, Cloud Platforms, Data Loading from Internal & External Sources, Docker, REST APIs, FastAPI, RabbitMQ, Python Automation

Reporting & BI: Power BI, Tableau, Data Visualization, Operational Reporting, Dashboard Design, Statistical Analysis, R, scikit-learn

Other: Production Support, Troubleshooting, Business Requirements Analysis, Stakeholder Communication, Agile/Scrum, Git, Linux/Bash